

Deterministic Optimization

Large-Scale Optimization

Dantzig-Wolfe
Decomposition

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Dantzig-Wolfe Decomposition 2

Large-Scale Optimization

Dantzig-Wolfe

Learning Objectives for this lesson

- Further develop the Dantzig-Wolfe decomposition algorithm for coordination and distributed optimization

Dantzig-Wolfe Decomposition

- The transformed problem is:

$$\begin{aligned} \min_{\lambda_1, \dots, \lambda_N} \quad & \sum_{i=1}^N \lambda_i (\mathbf{c}^T \mathbf{x}^i) \\ \text{s.t.} \quad & \sum_{i=1}^N \lambda_i (\mathbf{D} \mathbf{x}^i) = \mathbf{b}_0 \\ & \sum_{i=1}^N \lambda_i = 1 \\ & \lambda_i \geq 0, \forall i = 1, \dots, N. \end{aligned}$$

- N is the number of extreme points of $P = \{x : Fx = b, x \geq 0\}$

Example

- The number of extreme points of a polytope can be HUGE
- Consider a box in R^2 . It has 4 extreme points 
- A box in R^3 has 8 extreme points 
- A box in R^{10} has $2^{10} = 1024$ extreme points
- A box in R^n has 2^n extreme points!

Dantzig-Wolfe Decomposition

- Transformed problem has many variables, i.e. columns
 - Not many constraints

$$\begin{aligned} (MP) \quad & \min_{\lambda_1, \dots, \lambda_N} \sum_{i=1}^N \lambda_i (\mathbf{c}^T \mathbf{x}^i) \\ & \text{s.t.} \quad \sum_{i=1}^N \lambda_i (\mathbf{D} \mathbf{x}^i) = \mathbf{b}_0 \\ & \quad \sum_{i=1}^N \lambda_i = 1 \\ & \quad \lambda_i \geq 0, \forall i = 1, \dots, N. \end{aligned}$$

$$\begin{aligned} (MP) \quad & \min_{\lambda_1, \dots, \lambda_N} \begin{bmatrix} \mathbf{c}^T \mathbf{x}^1 & \mathbf{c}^T \mathbf{x}^2 & \dots & \mathbf{c}^T \mathbf{x}^N \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_N \end{bmatrix} \\ & \text{s.t.} \quad \begin{bmatrix} \mathbf{D} \mathbf{x}^1 & \mathbf{D} \mathbf{x}^2 & \dots & \mathbf{D} \mathbf{x}^N \\ 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_N \end{bmatrix} = \begin{bmatrix} \mathbf{b}_0 \\ 1 \end{bmatrix} \\ & \quad \lambda_i \geq 0, \forall i = 1, \dots, N. \end{aligned}$$

Dantzig-Wolfe Decomposition

- Step 4: Apply Column Generation to solve (MP)
 - Choose a subset I of columns. Solve the restricted master problem:

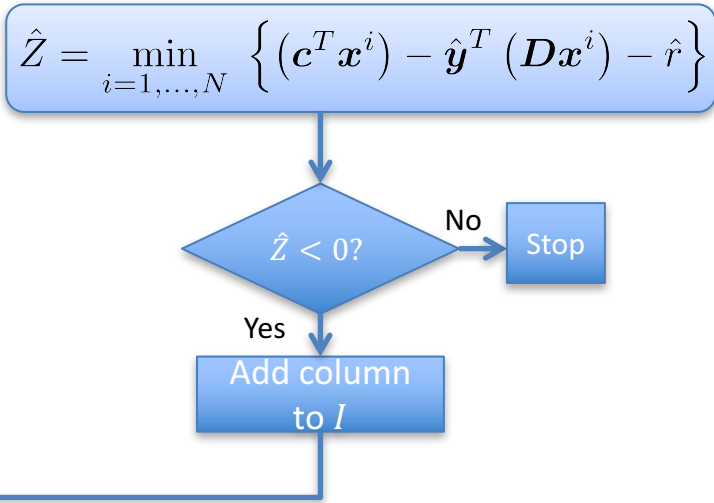
$$\begin{aligned} (RMP) \quad & \min_{\lambda_i, \forall i \in I} \quad \sum_{i \in I} \lambda_i (c^T x^i) \\ & \text{s.t.} \quad \sum_{i \in I} \lambda_i (Dx^i) = b_0 \quad \leftarrow \hat{y} \\ & \quad \quad \sum_{i \in I} \lambda_i = 1 \quad \leftarrow \hat{r} \\ & \quad \quad \lambda_i \geq 0, \forall i \in I. \end{aligned}$$

Obtain an optimal solution $\hat{\lambda}$, and optimal dual solution $(\hat{y}, \hat{r})$.

Dantzig-Wolfe Decomposition

- Step 4: Apply Column Generation to solve (MP)
 - Check optimality of $\hat{\lambda}$ using dual solution $\hat{y}, \hat{r}$:

$$\begin{aligned} (RMP) \quad & \min_{\lambda_1, \dots, \lambda_k} \begin{bmatrix} \mathbf{c}^T \mathbf{x}^1 & \mathbf{c}^T \mathbf{x}^2 & \dots & \mathbf{c}^T \mathbf{x}^k \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_k \end{bmatrix} \\ \text{s.t.} \quad & \begin{bmatrix} \mathbf{D}\mathbf{x}^1 & \mathbf{D}\mathbf{x}^2 & \dots & \mathbf{D}\mathbf{x}^k \\ 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_k \end{bmatrix} = \begin{bmatrix} \mathbf{b}_0 \\ 1 \end{bmatrix} \\ & \lambda_i \geq 0, \forall i \in I = \{1, \dots, k\}. \end{aligned}$$



Summary

- We learned:
 - The fourth step in Dantzig-Wolfe decomposition that applies column generation to solve the transformed master problem